

Flexible Strain Sensor Array

---EXP2-3 Safety in Numbers Application



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Introduction

The resistance (R) of the fibre is measured in both unstretched and 50% stretched states to evaluate its conductivity behavior. This screening mechanism is conducive to determining the generation of better electrical performance to meet the design requirements for applications such as flexible electronics or strain sensors. This poster mainly designs the application of flexible strain sensor arrays.

Property

High resistivity

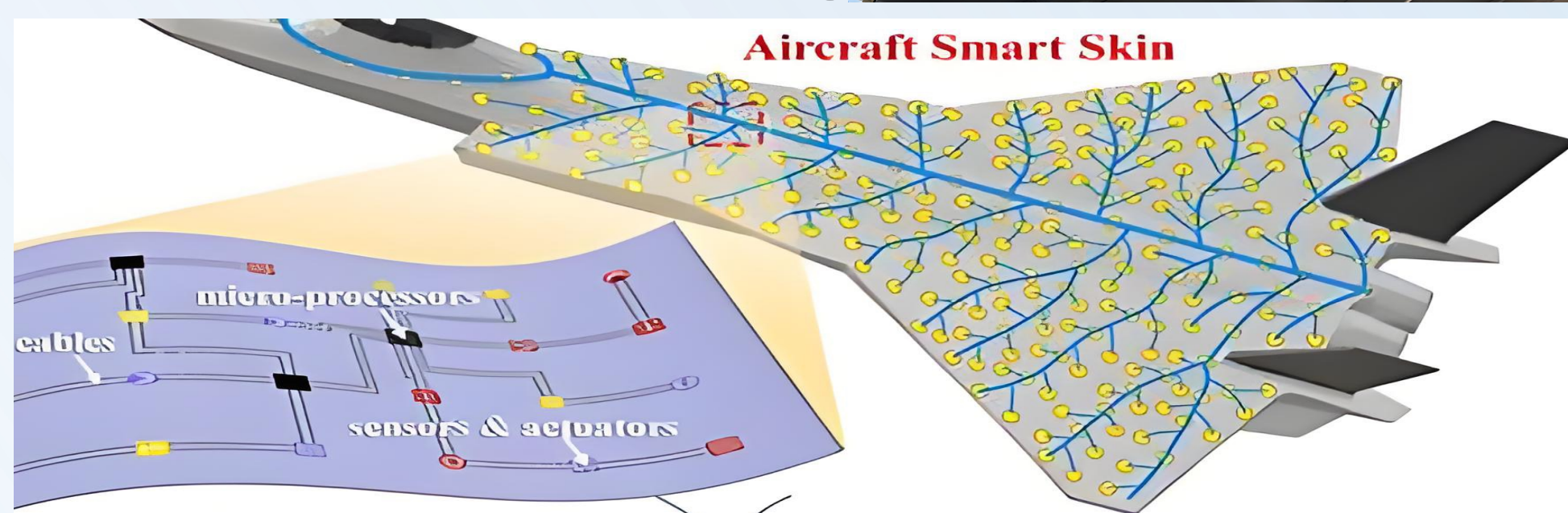
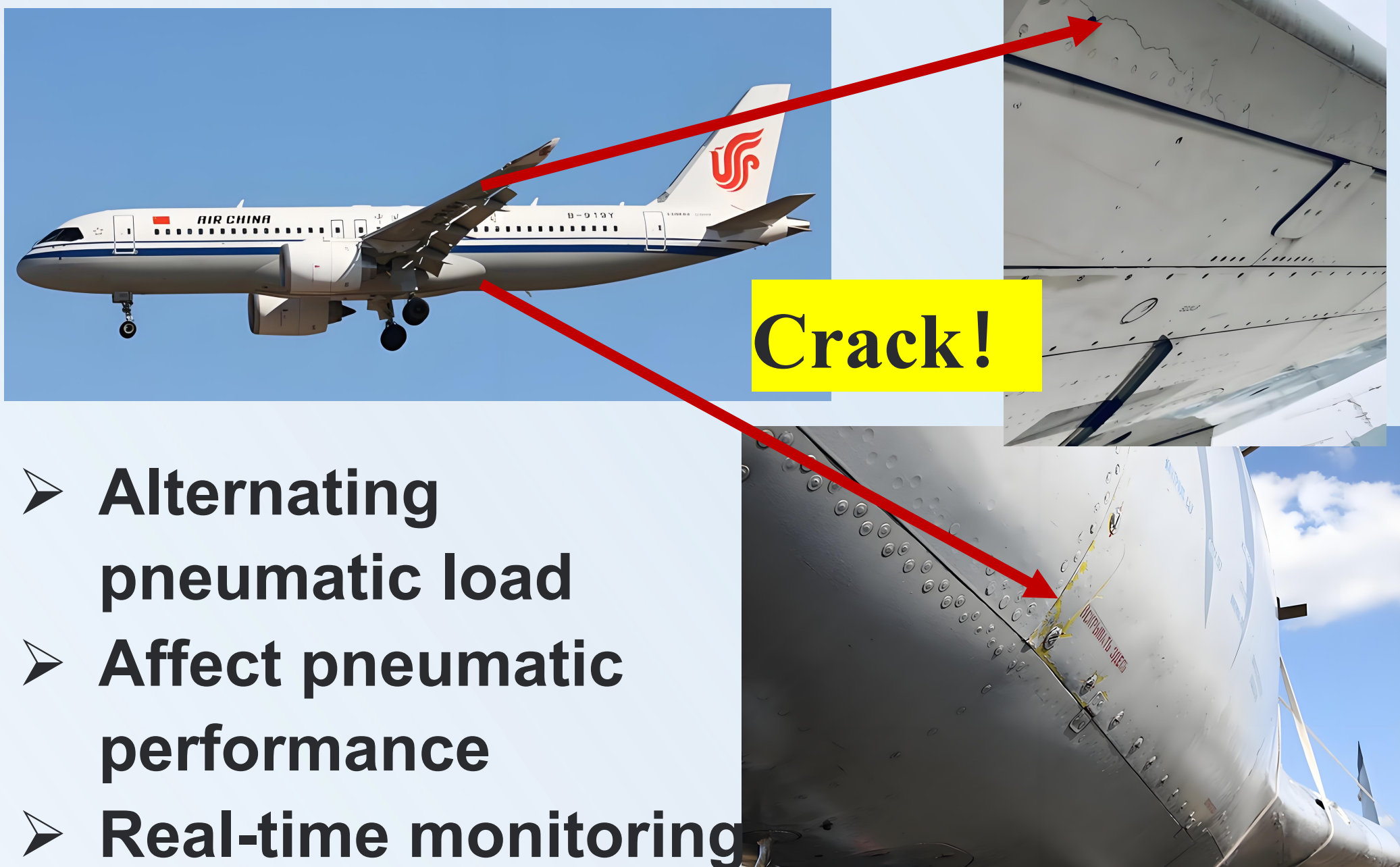


50% stretched --Higher R

sensitivity

reproducibility

Application



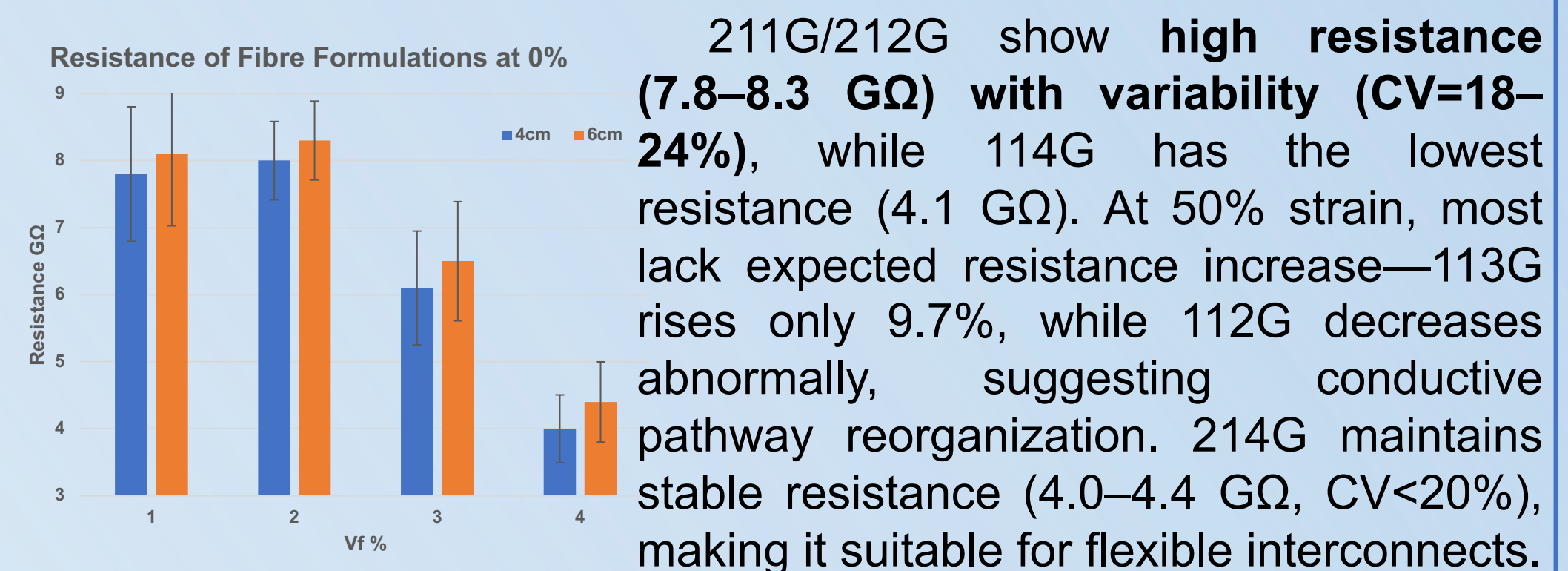
When a structure deforms due to mechanical stress, the conductive fibre embedded within it experiences a change in shape. This deformation alters its **electrical resistance or current flow**, which can be **monitored in real-time** to assess structural health.^[1]

This method allows for continuous monitoring, without the need for bulky sensors or manual inspections.

Signal Change = Structural Strain

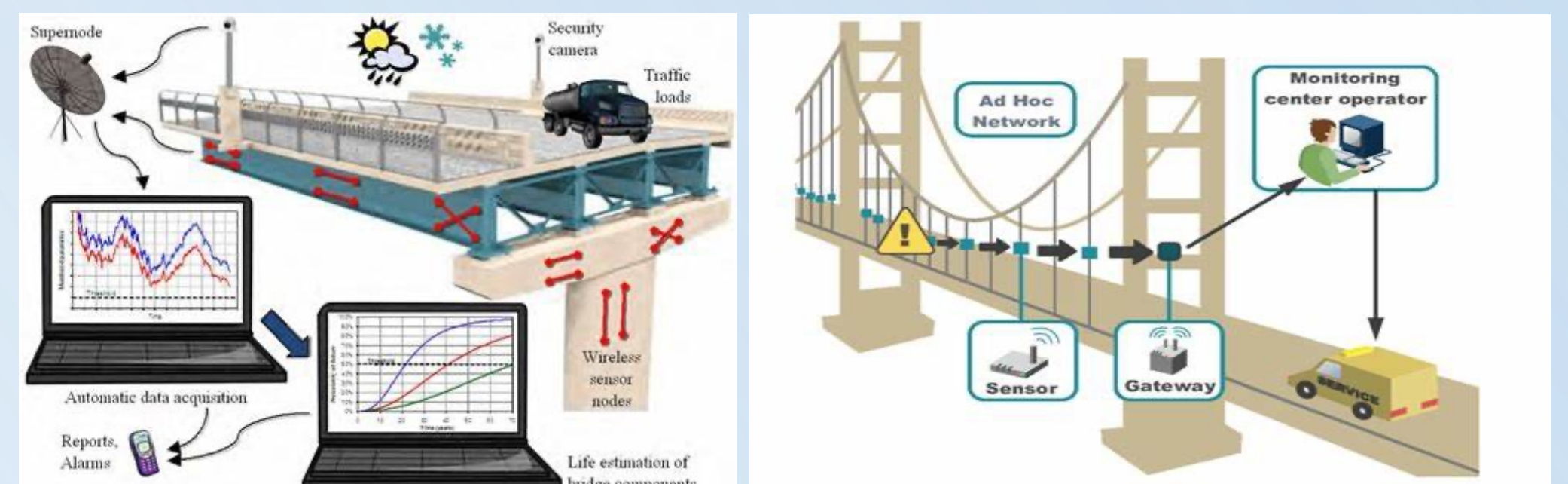
Example: 211G

4cm	Trail 1	Trail 2	Trail 3	Trail 4	Trail 5	Trail 6	Trail 7	Trail 8	Trail 9	Trail 10
211G/GΩ	8	9	10	8	6	5	7	8	7	10
6cm	Trail 1	Trail 2	Trail 3	Trail 4	Trail 5	Trail 6	Trail 7	Trail 8	Trail 9	Trail 10
211G/GΩ	8	10	10	8	6	5	8	9	7	10
Item	Mean	SD	Uncertainty	CV						
4cm	7.8	1.62	1.00	0.21						
6cm	8.1	1.73	1.071	0.21						

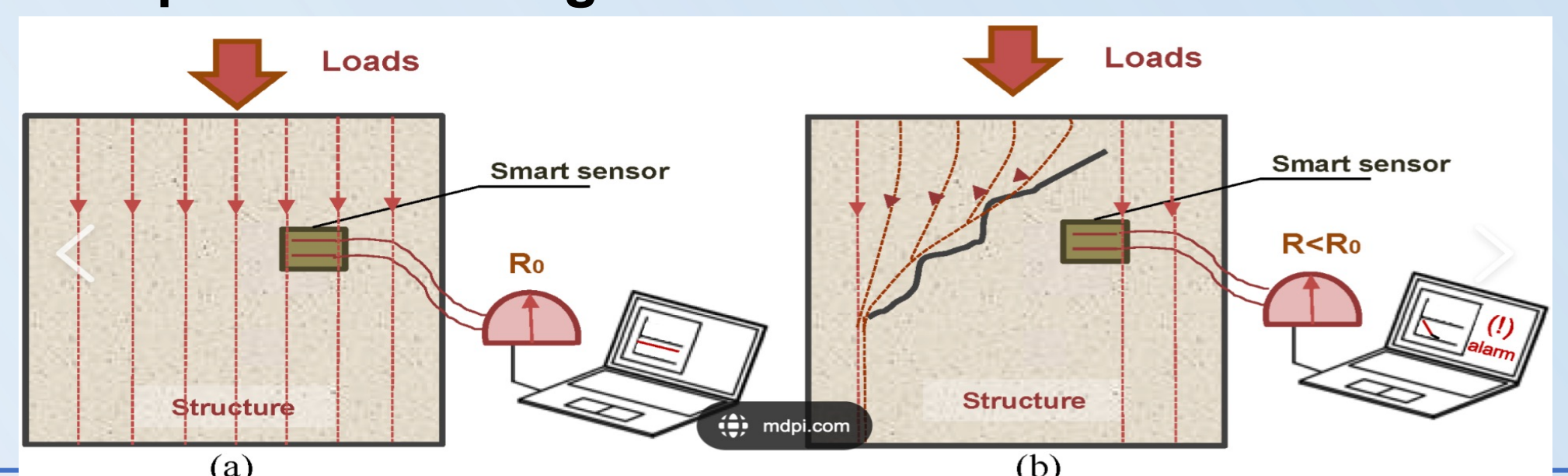


Application

For bridge: Deformation and Cracks



Monitor long-term deformation, vibration, or potential collapse risks in large-scale infrastructures.



Conclusion

Our ultra-fine conductive fibre is a **powerful enabler for future-proof structural monitoring**. By combining high sensitivity, durability, and compatibility with flexible materials, it presents a cutting-edge solution for safety-critical systems. Whether in the skies or across river spans, it provides **peace of mind through real-time awareness**.